

PRD: Pickleball Smash — Mobile Real-Time PvP

Version: 1.1
Date: May 31, 2026
Review Status:  Reviewed by reasoning agent — issues applied
Platform: Android & iOS (Godot 4)
Studio: Straw Hat Studio

1. Executive Summary

Pickleball Smash is a **mobile-first, real-time multiplayer pickleball game** built in Godot 4. It targets the massive gap in the pickleball gaming market — there are only 3 poor-quality pickleball games on mobile (avg rating ★2.5), while the sport itself has 50M+ players globally and is the fastest-growing sport in America. Meanwhile, Tennis Clash (★4.65, 471K ratings) proves massive demand for mobile racket sports.

Tagline: *"The fastest-growing sport, now in your pocket."*

2. Market Analysis

Competitive Landscape

Game	Rating	Reviews	Type	Notes
Pickleball Stars	★3.1	23	3D Sim	727MB, loot boxes, only real sim
Ultimate PickleBall	★2.3	45	Casual	Tournament style, 200 opponents
PickleBall 3D	★2.0	3	3D	New, buggy, claims MP
Tennis Clash	★4.65	471K	PvP Sports	Gold standard for mobile racket sports
Ping Pong Fury	★4.72	37K	PvP Sports	Swipe controls, online MP

Market Gap

- **Zero high-quality pickleball games** exist on mobile
- Pickleball's unique mechanics (kitchen zone, underhand serve, dinking, doubles) are unrepresented in any game
- No game has cracked **real-time PvP** for pickleball
- The sport's player base is growing 15-20% YoY globally

Target Audience

- **Primary:** Casual mobile gamers aged 25-55 (aligns with real pickleball demographics)
- **Secondary:** Pickleball enthusiasts who want to play on-the-go
- **Tertiary:** Competitive mobile gamers seeking a new PvP sports title

3. Game Overview

Core Concept

Real-time 1v1 and 2v2 pickleball matches with intuitive swipe controls, physics-based ball mechanics, and progression systems. Players compete in ranked matchmaking, tournaments, and casual games.

USPs (Unique Selling Points)

1. **Authentic pickleball physics** — Kitchen/no-volley zone, two-bounce rule, underhand serves
2. **Real-time PvP** — Matchmake against players worldwide (like Tennis Clash)
3. **Doubles mode** — 2v2 with AI partner or real friend (unique vs Tennis Clash)
4. **Accessible controls** — Swipe to hit, tap to position, drag for spin
5. **Growth system** — Ranked tiers, paddle/gear unlocks, cosmetic customization

4. Core Gameplay

Match Mechanics

- **Standard pickleball rules:** Underhand serve, must bounce once on each side (two-bounce rule), no volleys in kitchen zone
- **Scoring:** Rally scoring (every rally scores a point — faster, more casual-friendly). Traditional scoring (only server scores) considered for post-launch update
- **Game length:** First to 11 points, win by 2
- **Match types:** Singles (1v1), Doubles (2v2 with AI partner), Doubles (2v2 co-op with friend)

Controls

Input	Action
Swipe up	Lob shot (deep return)
Swipe down	Dink shot (soft, kitchen zone)
Swipe left/right	Cross-court shot
Tap + drag (aim)	Directional precision shot
Quick tap (before ball crosses net)	Volley

Input	Action
Hold + release	Power shot (risk/reward — easy to miss)

Shot Types

1. **Dink** — Soft shot landing in kitchen, resets the point
2. **Drive** — Powerful groundstroke, pace-heavy
3. **Lob** — High arc over opponent's head
4. **Volley** — Hit before ball bounces (cannot hit from kitchen)
5. **Erne** — Advanced: jump around kitchen to volley (unlockable skill)
6. **ATP (Around The Post)** — Advanced: hit around net post (unlockable skill)

Court & Visuals

- Third-person 3D perspective (behind and above player, like Tennis Clash/real sports games)
- Bright, colorful courts with customization options
- Ball trail effects for spin visibility
- Character avatars with emotes/celebrations

5. Game Modes

Core Modes

1. **Quick Match** — Casual 1v1, no stakes, fast matchmaking
2. **Ranked** — Competitive 1v1 with ELO/MMR system, seasons, rewards
3. **Doubles** — 1v2 with AI partner, or 2v2 with friend (co-op matchmaking)

Secondary Modes

1. **Tournaments** — 8/16/32 player brackets, running every few hours
2. **Challenges** — Daily and weekly skill challenges (e.g., "Win 5 points via dinks")
3. **Practice** — Solo court vs AI with adjustable difficulty levels. **Works offline** — uses cached AI profiles, no internet required
4. **Replay Mode** — Watch your last 5 matches. Works offline.
5. **Events** — Limited-time themed events (holiday courts, special rules)

6. Progression & Economy

Player Progression

System	Description
Player Level	XP earned from matches. Unlocks features, modes, cosmetics
Ranked Tier	Bronze → Silver → Gold → Platinum → Diamond → Elite → Champion
Paddle Level	Upgrade paddles for stat boosts (power, control, spin, reach)
Character Level	Each character levels up, unlocking unique emotes/skins

Monetization (Free-to-Play, Fair)

Item	Type	Notes
Paddle Skins	Cosmetic	Visual-only, no stat changes
Character Skins	Cosmetic	Outfits, court gear
Court Themes	Cosmetic	Different court colors/designs
Emotes	Cosmetic	Celebration animations
XP Boosters	Consumable	Double XP for limited matches
Season Pass	Premium Track	Tiered rewards, exclusive cosmetics
Paddle Sidegrades	Earnable	Trade-offs (power vs control, spin vs reach) — never pure stat upgrades. All earnable through gameplay

No pay-to-win. Paddles are **cosmetic-only** or **sidegrades** (trade power for control). No stat can be purchased with real money that gives a competitive advantage. All gameplay-affecting items are earnable through matches. Monetization follows Fortnite/Brawl Stars model: cosmetics, season passes, and convenience items only.

Currencies

Currency	Earned By	Used For
Coins	Matches, daily rewards	Paddle upgrades, basic cosmetics
Gems	Ranked rewards, purchase	Premium cosmetics, season pass, boosters
Tokens	Events, tournaments	Limited-time event cosmetics

7. Technical Requirements

Platform

- **Engine:** Godot 4
- **Primary:** iOS & Android
- **Potential:** Web (HTML5 export) for cross-platform

Online Requirements

Real-time matchmaking — WebSocket-based server (Nakama or custom Godot server)

- **Server-authoritative model:** Server validates all shot trajectories, timings, and ball physics. Clients send only inputs; server broadcasts verified state.
- Target 20-30Hz update rate with client-side prediction and position interpolation to smooth network jitter
- **Reconnection system:** 30-second reconnection window — AI takes over temporarily if player drops. Prevents frustration from mobile network switches
- **Lobby system** — Friend invites, party up before matches
- **Leaderboard** — Global + friends leaderboards

Performance Targets

- **iOS:** iPhone X and above, 60fps
- **Android:** 4GB RAM and above, 60fps
- **Install size:** < 150MB (significantly smaller than Pickleball Stars' 727MB)
- **Battery efficient** — matches under 15 minutes

8. Milestones & Timeline

 **Timeline estimate:** For a small team of 2-3 developers. Solo dev should double these estimates.

Phase	Scope	Duration
P0 — Foundation	Project setup, 2D physics engine, ball drag/aerodynamics, basic court rendering, touch input system, tutorial scene	3-4 weeks
P1 — Core Gameplay	Single-player vs AI, full court, all swipe controls, shot types (dink/drive/lob/volley), pickleball rules engine, gesture guide	6-8 weeks
P2 — Multiplayer	Server setup (Nakama), real-time PvP matchmaking, client-side prediction, position interpolation, reconnection system, basic ranking	6-8 weeks
P3 — Progression	Player levels, paddle sidegrades, currencies (coins/gems), shop, ranked tiers, season pass	4-5 weeks

Phase	Scope	Duration
P4 — Features	Doubles mode (AI partner), tournaments, daily/weekly challenges, events	5-6 weeks
P5 — Polish & Soft Launch	Animations, VFX, UI polish, audio, optimization, iOS/Android build, soft launch in 1-2 markets	4-6 weeks
P6 — Global Launch	Marketing materials, app store submission, localization (EN), live ops setup, server scaling	3-4 weeks

Estimated: 31-41 weeks (8-10 months) for a team of 2-3.

Scope cuts if timeline is critical (v1.0 only):

- Drop Doubles mode → save 5-6 weeks (move to v1.1)
- Drop Tournaments → save 3-4 weeks (move to v1.2)
- Drop Replay/Spectate → save 2-3 weeks (move to v1.3)
- **Minimum viable v1.0:** 1v1 Quick Match + Ranked + Practice + Shop = ~22-26 weeks

9. Success Metrics

Metric	Target (6 months post-launch)
Downloads	500K+
DAU	25K+
D1 Retention	40%+
D7 Retention	20%+
D30 Retention	8%+
Avg Session Length	8-12 min
Rating	4.5+
Revenue	TBD based on UA spend

10. Risks & Mitigations

Risk	Mitigation
Real-time PvP is complex to implement	Start with AI opponent in P1, proven P2P fallback for private lobbies
Godot's multiplayer ecosystem less mature than Unity	Use Nakama or dedicated WebSocket server; Godot 4's ENet/MultiplayerAPI is adequate for authoritative server model
Small team for 2v2 real-time sync	Phase 4 after 1v1 is stable; simplify with AI partner controlling one side
	Localized marketing, target US/UK/Canada first

Risk	Mitigation
Pickleball niche may be too small in some regions	
Server cost for authoritative physics calculations	Use lightweight server (single binary) with state validation only; actual physics runs client-side with server verification